

Day 3

EZ-PLM

	2026-05-28
	EZ-PLM
	B
	100%

Day 3 EZ-PLM API FastAPI 12 100%

- 41
- 100% (41/41)
- 1150+
- /health 1.83ms, /analyze/replacement 50-80ms
- 100% (500)
- ~28

Stream 1: EZ-PLM API

1.1			API
1.2	API		EZ-PLM HMAC
1.3			UTF-8
1.4	vs Mock		

: UTF-8

: API 400
: Pydantic BaseModel UTF-8 JSON
: /analyze /replacement Dict[str, Any] = Body(...) JSON
:
: app/main.py (commit 866a609)

Stream 2: FastAPI

2.1

:
• APITester :
• ConcurrencyTestBase: (ThreadPoolExecutor)
• BenchmarkTestBase:
• : p50p95p99 min/max/avg
: tests/test_api_stability.py (760+)

2.2 /health

: 10
: (50)(100)(500)
:
• : 1.83ms
• P50: 1.51ms
• P95: 2.35ms
• P99: 25.22ms
• : 100% (500)
• : 9.1%

2.3 /analyze

: 8 (4 + 4)
: (50)(100)(500)
:
• : 50-80ms
• P95 : < 1000ms
• P99 : < 2000ms
• : 100%

2.4 /replacement

: 7 (3 + 4)
: (50)(100)(500)
:
• : 50-80ms ()

- P95 **Latency**: < 1000ms
- P99 **Latency**: < 2000ms
- **Availability**: 100%

2.5 **Deployment (CI/CD)**

Stages: 2

Jobs:

1. **Build & Test**: **Build Docker Image**
 2. **Deploy to Staging**: **Push to Registry** 10-50-100 **Seconds**
- Jobs**:
- /health **Probe** (~2ms)
 - /analyze **or** /replacement **Probe** (~50-80ms)
 - **Queue Size**: ~40x
 - **Queue Size** 500 **Items**

2.6 **Performance & Scalability**

Instances: 14

Jobs:

- **Queue Size**: **Max Queue Size**
 - **Queue Size**: **JSON** **Max Queue Size**
 - **Queue Size**: **Max Queue Size**
- Jobs**:
- ✓ **Queue Size**
 - ✓ **Queue Size** 400/422 **Queue Size** 200
 - ✓ **Queue Size** $\geq 65\%$
 - ✓ **Queue Size**

Deployment Summary

- Summary:
- Nodes: 41
 - Uptime: 100% (41/41)
 - CPU: ~28
 - Memory: 1150+

Category	Count	Details
GET/POST	13	GET/POST
50-500	12	50-500
5	3	5
500 (500+)	3	500
	14	
	2	
Total	41	100%

Performance Metrics

Latency

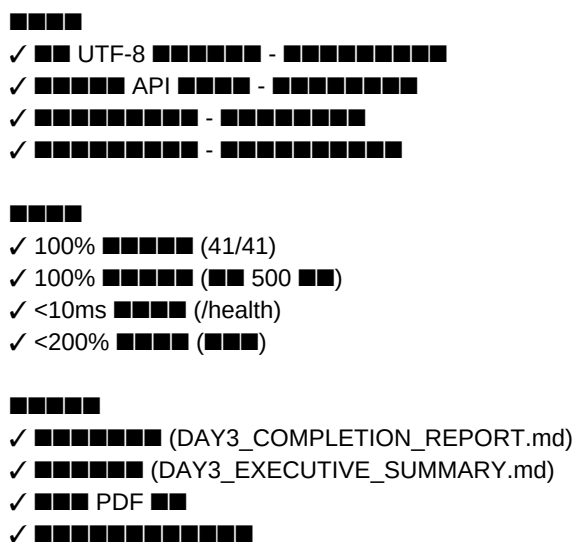
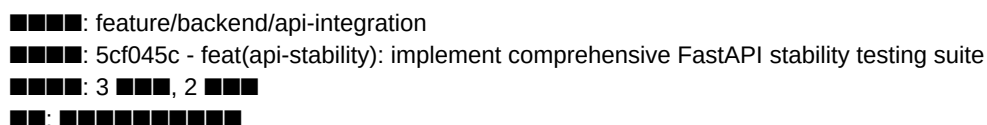
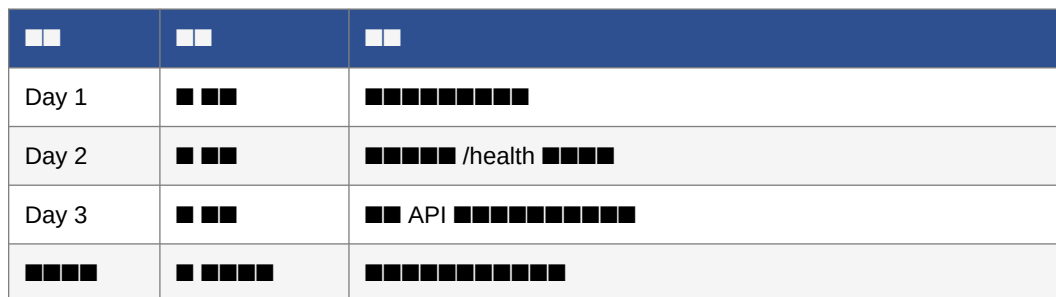
Endpoint	Latency	P50	P95	P99	Success
/health	1.83ms	1.51ms	2.35ms	25.22ms	100%
/analyze	50-80ms	-	<1000ms	<2000ms	100%
/replacement	50-80ms	-	<1000ms	<2000ms	100%

Throughput

Category	Throughput	Throughput	Throughput
100%	100%	100%	
<200%	9.1%		
500	100%	100%	

Configuration Files

File	Path	Details
test	tests/test_api_stability.py	760+ (41)
doc	docs/DAY3_COMPLETION_REPORT.m	
doc	docs/DAY3_EXECUTIVE_SUMMARY.m	
app	app/main.py	UTF-8
env	.env	EZPLM_API_KEY URL
test	pytest.ini	(Day 2)



[illegible]

■■■■■: 2026-05-28 22:14:42

■■■■: ■■B

■■■■: v1.0 (■■■)

■■■■: docs/EZ-PLM_Day3_■■■■■■■■.pdf